

Weaving ML with Human Aesthetic Assessments to Augment Design Space Exploration: An Automotive Wheel Design Case Study

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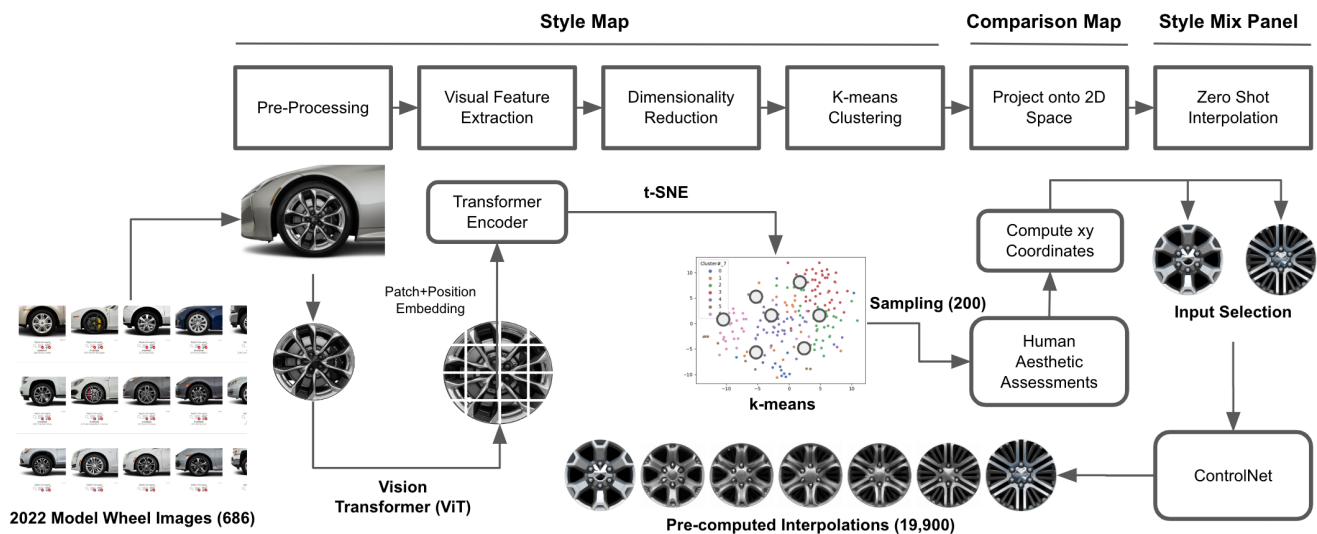


Figure 1: ML Pipeline for Design Space Organization System.

ABSTRACT

Consumers’ emotional and cognitive attachment to product design plays a pivotal role in influencing purchasing choices. Therefore,

*This work was completed during an internship.

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product designers incorporate this signal as they develop new products. The goal of our work is to reduce the psychological distance between designers and consumers in the automotive concept design process. While generative AI models hold the potential to amplify creativity, these models do not have any of this specialized knowledge. In this work, we developed a novel framework and system that combines machine learning, human aesthetic assessments, and visualization to support designers in organizing a large space of automotive wheel designs. We present a case study with 10 automotive designers using the tool to inspire novel wheel designs and end with a discussion of use cases and design implications for using this framework to support professional product design practice.

CCS CONCEPTS

• **Applied computing** → *Media arts*; • **Human-centered computing** → *Interaction paradigms*.

KEYWORDS

Generative AI, Interpolation, Automotive Wheel Design

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1 INTRODUCTION

People’s emotional connection with product design is an important signal that drives purchase decisions and overall satisfaction. This relationship can be characterized by a set of semantic aesthetic attributes (e.g., rugged, sporty, luxury) assigned to perceptual features of a particular design. In the concept design phase, capturing and responding to this signal is an important part of a product designer’s job. Yet in automotive design, where online experimentation is not a viable option, it is driven by speculation about consumers’ aesthetic preferences, drawing from designers’ intuition, prior experience, and domain knowledge. In this process, designers search and curate a set of potential design examples from experience and online repositories such as a product catalog or Pinterest in response to a design prompt (e.g., “a rugged, adventure seeking vehicle”), to establish a design direction and inspire a generation of new design alternatives that best fit the design prompt.

We identify two problems that exist in this process to address through a novel framework that combines machine learning (ML), human assessments, and interface design. First, it does not consider a systematic exploration and investigation of design choices with respect to different semantic attributes that represent perspectives of existing and prospective consumers [9]. Additionally, consulting direct design feedback in later production phases is too costly. In the automotive industry, targeted consumers are recruited to make aesthetic assessments in design clinic sessions based on descriptors such as sporty and luxurious. A single design clinic for a new vehicle design could cost over \$100,000, with annual expenditures estimated at tens of millions of dollars for a single manufacturer [3]. These problems are further compounded by additional design iteration cycles to address important customer feedback that cause potential delays in productization.

In our work, we aim to reduce the psychological distance between designers and consumers and increase designers’ capacity to make sense of a large design space. To this end, we introduce a creativity support tool (CST) that incorporates ML techniques to support designers in organizing a large number of automotive wheel designs informed by human aesthetic assessments, enabling flexible and creative modes of thinking about novel designs. We report on four use cases that demonstrate how this CST can facilitate (1) rapid ideation and sketching, (2) validation of aesthetic

assessments, (3) finding happy accidents in form studies, and (4) visual communication in automotive design practice.

2 CHALLENGES IN APPLYING GENERATIVE AI TO PRODUCT DESIGN

Fueled by the emergence of AI-based text-to-image generation tools, such as Midjourney¹ and DALL-E², various design professions, including, but not limited to, graphic design, film making, interior design and creative ad agencies are increasingly experimenting with new ways of translating abstract concepts into concrete visual representations, to support their practice [17]. While enticing, current generative systems have limited applicability in professional industrial product design for two reasons: (1) designers lack the means to anticipate outcomes based on text input, and (2) it is largely unknown whose aesthetic assessments are represented in the design outcomes.

Recent prior work in ML and Human-Computer Interaction (HCI) has demonstrated the feasibility of combining text-image joint embeddings and image generation models to achieve a sufficient approximation of visual style based on a given concept keyword. For instance, StyleCLIP [13] utilizes a CLIP model [15] to enable interactive text-based image manipulation in StyleGAN-generated images by recognizing meaningful semantic changes in latent representations. 3DALL-E [11] is a plugin within a CAD software that combines DALL-E [16], GPT-3 [2], and CLIP [15] to help users interactively guide 3D model development from text input. Despite these exciting advances, designers’ may struggle to fully convey their intents in text, thus leading to poor control and precision over the generated outcomes [5].

Recent advances in Large language Models (LLM) and text-image joint embeddings gave rise to novel ways of eliciting and estimating human aesthetic assessments. Vast consumer data are expressed and captured as unstructured text, behavioral, and physiological signals, and analyzed to create personas that represent consumers’ demographic, behavior, and preference profiles [8]. Personas then help ground design in the context of consumer data and play a significant role in guiding the design process. Indeed leveraging textual representations of consumer preferences is valuable, but there is potential downside to relying on LLM-based personas or virtual audiences³ to stand in for a specific target demographic. We lack a complete understanding of whose viewpoints are encapsulated in the generated responses, which carries the risk of either over- or under-representing certain demographic subgroups [18]. Consequently, designing new products directly informed by the combined application of text-image joint embeddings and image generation models could result in products that target an extreme niche group or fail to align with the values and preferences of the intended consumer group. Moreover, consumers are changing in a 3 to 5-year time horizon, a challenging prospect that adds considerable cost burden and uncertainty in *getting the design right* [4].

Our paper offers three contributions in response to these challenges: 1) a novel framework that combines interface design, ML,

¹<https://docs.midjourney.com/>

²<https://openai.com/dall-e-3>

³<https://www.syntheticusers.com/>

Table 1: Mapping of different ideation phases to goals and features of Design Space Organization (DSO)

Ideation Phase	Requirements	Design Goals	Features
Data collection	Exploration	Support efficient and flexible exploration of a large number of design examples	Style map
Mapping and consensus building	Analysis	Project designs onto 2D space of consumers' semantic interpretation	Comparison map
Design elaboration	Generation	Provide efficient means to suggest design alternatives within users' expectations	Style mixing panel

and human aesthetic assessments, 2) an AI-based prototype system that facilitates exploration of various design alternatives, while incorporating the consumer's interpretation of semantic expressions, and 3) an early investigation of the CST's applicability in the real world design setting.

3 FORMATIVE STUDY

To better understand the context through which ML could assist product design and development, we conducted a formative interview study with professional exterior designers in a large automotive manufacturing company. We recruited and interviewed five practicing design professionals who are also subject matter experts⁴ ($\mu=17.4\text{yrs}$, $\sigma=6.4\text{yrs}$) in the wheel design process. During the interview, we asked about their current ideation practices and barriers that hinder creative activities. We also asked about different styling keywords that they use in practice, which helped us identify a set of 25 style keywords (see Appendix B.1) to inform subsequent data collection. Below, we summarize the wheel design ideation process, current challenges, and possible solutions that emerged from the interviews. When reporting interview quotes, we use P_f^X to denote participant number X in the formative study.

First, in the **data collection** phase, designers spend significant time and resources in researching reference images used in benchmarking to develop insight into design trends in the current market and establish an aesthetic baseline. Due to the sheer number of potential reference designs that could be collected from competitor brands, however, designers lamented about challenges in reviewing all available design examples and opted for a focused search within the target market segment (e.g., particular grade of trucks such as off-roading).

Moreover, the efficiency gained from image search engines such as Google and Pinterest could come at the cost of design fixation, a cognitive bias that is detrimental to the creative design process [10]. For example, designers may become fixated on Pinterest's algorithm-driven recommendations, and struggle to diverge from learned patterns.

In the **mapping and consensus building** phase, designers review and hold regular discussions about how particular design examples fit into a 2-dimensional mapping space organized by xy-axes. They typically assign semantic keywords to each axis, of which one axis represents a continuum between two extremes of a particular semantic dimension (e.g., low to high luxury). In

weekly design reviews, designers discuss adequate placement of different wheel design examples within this 2D mapping space while physically placing printed designs on a large wall.

In our conversations with designers, we learned that even designers have different understandings of key concept terms, and the mapping activity facilitates consensus building among them. P_f^1 commented on the need for doing so.

I would like to mention when we use the word 'luxury,' it can make people think of all sorts of different things. So, it's important to be clear and nail down what exactly we mean by it to avoid any mix-ups and make sure we're all on the same page.

Such concept mapping meetings would last up to six months where they later include other decision makers such as marketing, sales, engineering, and product planning departments who would also invest a significant amount of time to align their projections within the design space of wheels.

In the **design elaboration** phase, designers gradually establish the direction of design development through ongoing weekly design reviews where they generate multiple alternative designs. Yet this process for creating new style variations is mostly limited to simple visual style variations such as increasing the number of spokes⁵. Because designers bring their own reference images and derivative concept sketches to these meetings, cross pollinating design examples among different designers by way of merging two different wheel designs naturally increases the chance of adding meaningfully distinct style variations.

Communication in the design elaboration process, while important, can be abstract. For instance, designers are frequently advised to merge different wheel designs to increase style variation. Having more images helps counter the inefficiencies that come with communicating at abstract levels. While helpful, P_f^2 mentioned that generating new images for the sake of improving communication is counterintuitive and time-consuming.

When discussing design concepts with other designers or stakeholders, communicating only through words without images is too abstract. That's why we prepare a lot of images. However, it's not easy because creating images takes a lot of time.

⁴This is a small team of designers in a US-based design studio that is focused on designing automotive products in the North American market.

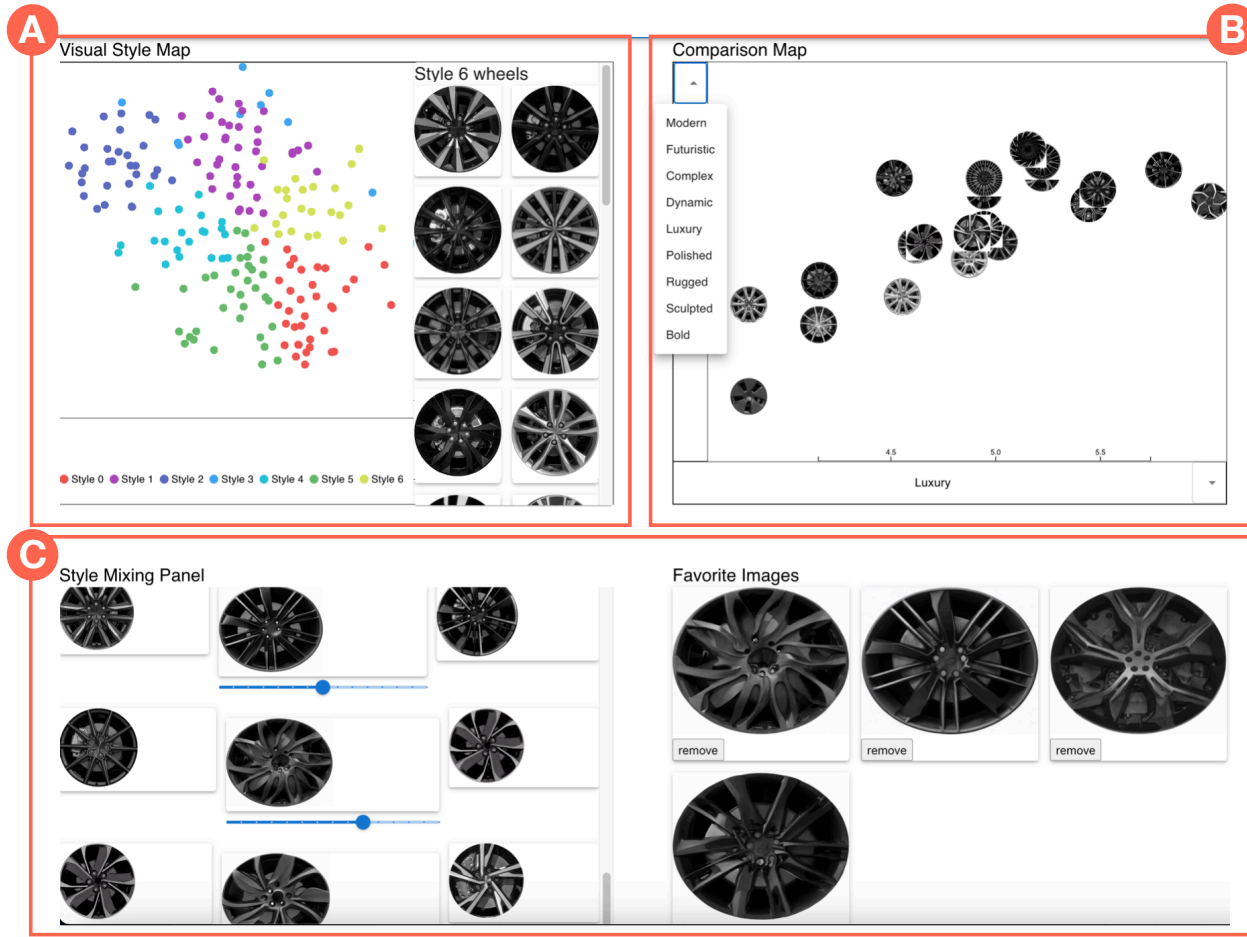


Figure 2: DSO with three main interactive visualization interfaces – Style map (A), Comparison map (B), and Style mixing panel (C). Style map (A) organizes design examples by similar visual features to reveal emerging design patterns in the market. Comparison map (B) helps designers understand consumers’ aesthetic assessments of wheel styles along particular semantic attributes. Style mixing panel (C) then enables visual style mixing experiments from two selected source images.

4 SYSTEM DESIGN

The insights we derived from the formative interviews directly informed the design of an AI-based CST system—Design Space Organization (DSO). In this section, we introduce our design goals, describe three main features of the user interface, and the ML and data pipeline that enable these features.

4.1 Design goals

From the three ideation phases identified through our formative study, we distilled relevant requirements and design goals that informed the design of the DSO system (Table 1). These goals include the following:

- Support efficient and flexible exploration of multiple design examples to encourage exploration in ways that expand the design space to include atypical categories.

- Project designs onto a two-dimensional space to support meaningful engagement with and analysis of consumers’ aesthetic assessments of said design examples.
- Provide fast and low-cost means to suggest design alternatives that are within users’ expectations.

4.2 DSO User Interface

DSO is an AI-based CST that allows designers to explore a large number of design examples and alternatives. The system provides important information for designers during the wheel styling process, including visualization of style information organized by visual features and representation of human aesthetic assessments of design examples. Using latent diffusion models, designers can build on this visual information to experiment with novel visual concepts by interpolating between two-wheel designs. In effect, the system aids designers in generating new alternative designs aligned with their intended design direction.

⁵A spoke refers to the rods radiating from the center of a wheel

DSO consists of three main features (Fig. 2). **Style map** (Fig. 2-A) provides a large number of design samples ($n=200$) in a two-dimensional cluster map organized by visual features such as the number or thickness of spokes in a wheel. Each cluster group corresponds to a distinct style group within the Style map. Users can explore different style groups by clicking on each colored data point, which updates a 2-column image grid of wheel images belonging to the particular style group. Style selections are displayed on a 2D spatial layout, or **Comparison map** (Fig. 2-B), based on quantitative human ratings (on a 7-point Likert scale) of images in association with 9 semantic attributes. The placement of images changes according to the chosen semantic attribute in x/y axis and corresponding values. Selections made on the comparison map are displayed on the **Style mixing panel** (Fig. 2-C), where users can perform visual style mixing experiments using two selected input images. Users can drag the slider to explore interpolation results that are either close to or far from either input image. The results of interpolation can be used as input to generate diverse combinations of images for further experimentation and exploration.

5 ML AND DATA PIPELINE

Figure 1 shows the ML and data pipeline powering the three features of DSO. The following sections describe the individual steps we took to build these features.

5.1 Visual Feature Extraction and Clustering

We employed a Vision Transformer (ViT) [6] model to extract representative visual features from the set of 686 wheel images. We then used a t-SNE [20] dimensionality reduction algorithm to reduce the data to two dimensions. The next step involved determining an appropriate number of clusters (k) for K-means clustering [7]. We utilized the Elbow method [12] to identify these optimal values, in which the number of clusters is seven. Subsequently, we evaluated the distances between the cluster centroids and all data samples in each case and selected the 200 samples that were closest to their respective centroids across all cases.

5.2 Human Semantic Aesthetic Assessments

We identified a set of 25 style keywords through a formative study involving discussions with five automotive designers. We narrowed down the list to 9 style keywords after computing the CLIP scores associated with all 200 wheel images (see Appendix B.1. We filtered out keywords of low CLIP scores with wheel images as these keywords may not be commonly used to describe wheels by the general population. In total, we identified nine style keywords to be used in obtaining human aesthetic assessments: Modern, Futuristic, Complex, Dynamic, Luxury, Polished, Rugged, Sculpted, and Bold. We recruited 160 participants on Prolific to rate 30 images along the 9 semantic dimensions, resulting in each image receiving an average of approximately 15 ratings. This process allowed us to construct a dataset consisting of 200 images, each tied to average values of 15 ratings for each of the nine design dimensions.

5.3 Zero Shot Interpolation

We used an implementation of ControlNet that enables zero-shot controllable interpolation [21] with latent diffusion models to support designers in generating and exploring multiple design alternatives in a focused design space. This implementation allowed us to compute 15 interpolation slices between any pair of input wheel images. The interpolation model takes three conditional inputs: (1) a pair of input images, (2) text prompts, and (3) edges extracted from the input images using Canny edge detector [1]. We generated a total of 298,500 images by computing 15 interpolations between 19,900 unique combinations of wheel designs from 200 sampled images. These images are stored in an AWS server and retrieved via API calls when two-wheel images are selected in the Style Mixing Panel.

6 POTENTIAL USE CASES OF THE DSO SYSTEM

The goal of our exploratory user study was to determine whether and how DSO, a system that weaves ML with human aesthetic assessments, might support designers' creativity while discovering novel wheel design patterns. We recruited 10 automotive design professionals and students who are currently enrolled in transportation programs in industrial design schools and have some form of professional experience such as internships (see Appendix ??). Each study lasted no more than 2 hours and participants received a \$100 gift card for compensation.

Participants were asked to use DSO to complete three steps: (1) explore and review multiple wheel design examples, (2) examine the relationship between wheel designs and aesthetic ratings relative to each semantic dimension, and (3) generate five new wheel designs by selecting two input images. These steps were repeated in two-wheel design tasks that respectively focused on a sedan and truck body. An image and design prompt were provided in both tasks.

We uncovered insights that highlight potential use cases of this CST. We summarize these use cases below. When reporting interview quotes, we use P_u^X to denote participant number X in the user study.

6.1 Rapid Ideation and Sketching

Many participants in our study expressed excitement over the tool's ability to support rapid ideation and sketching. For instance, a 20-year veteran (P_u^1) suggested that the tool allows for quick benchmarking experiments with a diverse range of designs, enabling rapid ideation in a larger design space.

In a typical studio setting, we are only focused on a limited set of inspirational benchmarks, but that limits diversity. This tool allows designers to quickly combine and explore a diverse range of combinations that they might have not thought of.

While some designers approached the tool as a strategic decision making tool, other designers were more intuitive and emotional in their approach. A junior transportation design student P_u^8 shared his excitement over the way in which the tool accelerated the ideation process to a point that was aligned with the timing of his emotional reaction.

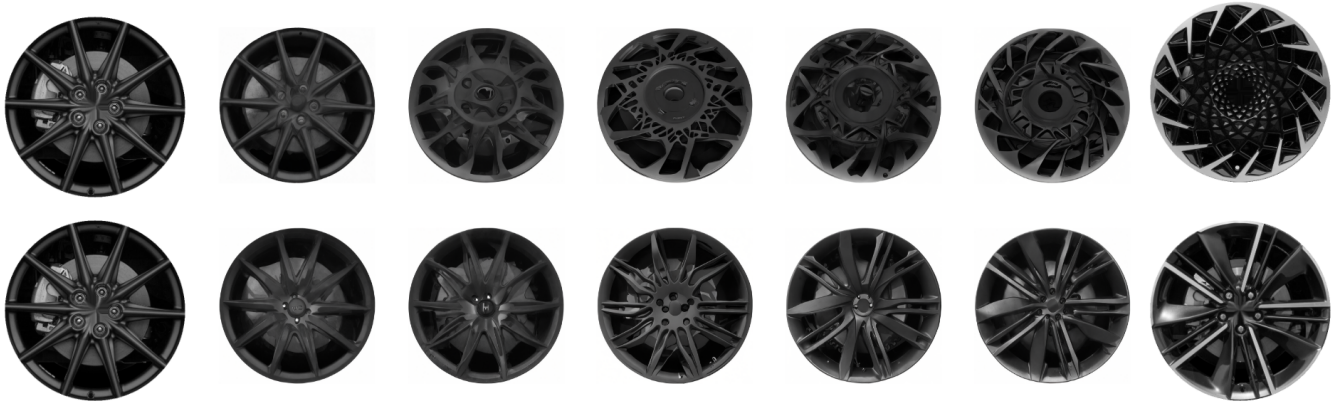


Figure 3: Comparison of interpolation results. Interpolating between completely different wheel styles resulted in poorer quality (Top), whereas compatible designs tend to yield higher quality results (Bottom).

I saw myself emotionally reacting to a wheel more quickly than sketching on my own—sometimes my hand is not putting down exactly what the brain is thinking.

Participants also mentioned a further use case to sketch on top of resulting inspirational images. Using an inspirational source image as an underlay for sketching is a common technique that is used among industrial designers. Doing so adds a layer of abstraction on which they can carry out visual experiments by adding lines that slightly deviate from the source design. Both P_u^5 and P_u^8 suggested that while the resulting interpolations generated by AI would not automate their work directly, they would serve as inspirations to support mental processing which designers then translate into quick sketches.

I can see myself using this in the real world. I'll use it for quick sketches, if my manager asks me to come up with five sketches for a car by next week, I'll use it as a tool to get basic ideas. (P_u^5)

I see myself using it as a way to quickly jot down initial ideas and utilizing those initial ideas to focus more on mental processing towards achieving the final result, rather than attempting to physically generate numerous ideations [...] It feels like it would save a significant amount of time. (P_u^8)

6.2 Validating Aesthetic Assessments

When asked about their experience using the Comparison Map, participants told us that they were able to develop confidence in their chosen design direction by consulting relative ratings each design example received on any two selected dimensions in the x- and y-axis and validate their own assessments in analyzing the data. P_u^2 expressed his excitement.

It was really helpful in understanding the consumer's perspective. I had quite a few thoughts that were different, and I believe that the ability to validate them is the biggest advantage of the Comparison Map.

However, the benefits that came with the simple representation of subjective human aesthetic assessments ran counter to the goal

of the Comparison Map. Strong alignment between designers and consumers' assessments boosted their confidence whereas weak alignment resulted in their dismissal of results of the Comparison Map. Some participants were eager to explore the data further and suggested increasing the transparency of the data to support their validation needs. P_u^3 illustrated this point very well.

In cases where the difference was substantial, I found my confidence in the data to be lower. As a possible solution, I believe it would be beneficial to have access to demographic information for consumers who rated the Comparison Map scores. If this data aligns with the persona I have in mind, it could significantly boost my confidence in the results, even in the presence of significant differences.

6.3 Discovering Happy Accidents in Form Studies

In industrial design, form studies consist of sequential, comprehensive manipulations of a geometric volume. Many participants in our study saw the value of this tool in assisting their study of different forms in a wheel style. While designers saw the benefits of seeing unexpected AI artifacts, calling it a *happy accident*, others also expressed disappointment. A junior level professional automotive designer (P_u^6) appreciated and found meaning in the imperfect interpolation results:

Getting surprising results from AI is a real win—it's like having AI mix opposite images and create something really cool. When I see the details I wanted, it's a total satisfaction boost!

On the other hand, P_u^5 felt dissatisfaction from the distortion in the interpolations.

The unexpected curves are an advantage of AI. Conversely, I felt disappointed by its apparent difficulty in creating straight-line designs.

These reactions could be attributed to designers' unrealistic expectations of AI as well as the overall lower interpolation quality between designs that are incompatible. In general, attempts to mix

two very different wheel styles resulted in large amounts of distortions in the interpolations (Fig. 3). In response, designers desired additional functionalities to directly edit the results of interpolations in ways that aligned with their expectations.

6.4 Visually-Aided Communication

Lastly, designers found immense value in using concrete design examples from the generated interpolation images and aesthetic assessments as means to support design communication with management and colleagues. For example, P_u^1 remarked on the value of aligning subjective interpretations to remove unnecessary inefficiencies in communication:

We often spend more time aligning our thoughts than we realize, and this feature could be applied to all design tasks [...] to support clear communication between design managers and designers.

A second year MFA transportation design student (P_u^3) commented on this topic as well, describing its potential value in supporting communication among multiple designers.

It would also be very helpful in designer-to-designer communication. Most discussions about design are highly abstract, but if we can quickly visualize what we want, it could save us a lot of time.

7 DESIGN IMPLICATIONS FOR AUTOMOTIVE PRODUCT DESIGN

In this case study, we developed and explored the applicability of a novel framework that combines ML, human aesthetic assessments and interactive visualization in the context of an automotive wheel design process. We now turn to summarize our lessons learned along with two design implications for automotive product design, including 1) aligning generative AI with intent expression to increase designers' capacity to control the outcomes of generative systems and adapt to new design practices with AI, and 2) embracing transparent data to improve the perceived value of human aesthetic assessments in design time.

7.1 How Much Control Between Designers and AI?

In further conversations with our participants who used mainstream text-to-image generation tools such as Midjourney, we learned that leaning too much on the stochastic properties of latent diffusion models rendered the output images impractical for real world use. Happy accidents may occur at infrequent intervals, but the unpredictability of the chances deters their use. In conceiving DSO, we hypothesized that providing designers more control early on to scope the boundary to a focused design space—by selecting two wheel styles they intend to mix together—would help capture their intended design expression.

Our study results suggest that designers feel the output of the interpolations to be surprising and creative when they could reasonably understand the boundaries of the design space. Within this narrowly defined design space determined by two selected input images did they find 'happy accidents' that are meaningful while still being predictable. We argue that there is merit and promising potential in initially constraining the influence of AI's role in design

and preserving designers' intentionality [5] as a suitable approach to help increase their capacity to adapt to new design practices with AI in the professional product design context. This insight also underscores the idea that Generative AI tool outcomes do not necessarily have to be perfect. By reframing our perspectives on distortions, artifacts, and hallucinations as features rather than flaws in creativity domains, we believe there will be renewed interest in the value that Generative AI can offer in product design.

7.2 What Role Can Human Aesthetic Assessments Play in Design Time?

Design is a wicked problem through which uncertainty becomes a currency of practice. It is the designer's job to tolerate and reduce a particular amount of uncertainty through interpretive exercises grounded in numerous discussions among them. Our case study has illustrated the value of visualizing aesthetic assessments in design time, which can lend support to designers engaging meaningfully with data and unlock their potential to manage uncertainty about what design solutions might capture their consumers' excitement. Our study results show that the authenticity of consumer opinions was a beneficial context in supporting designers' exploration of the design space. By comparing consumers' aesthetic assessments with their own, designers could increase confidence in their chosen design. However, we also observed differences in semantic interpretation between designers and consumers within the Comparison map. When these differences crossed the designers' threshold for tolerating discrepancies in the data, the feature appeared to have detrimental effects on their perceived value of consumer data.

Our goal is to provide opportunities and tools that minimize the 'psychological distance' [19] that designers experience as they analyze consumer aesthetic assessments in design time. Based on early feedback from few designers in our study, we are optimistic about the potential for increased data transparency and techniques to enhance interactive data exploration to play in design reviews, to add meaningful interpretive layers that facilitate data-driven decision-making processes. In one scenario, visualizing the dispersion and distribution of data to peruse polarizing aesthetic opinions could help designers better understand why their customer base has an extreme affinity or hatred toward a particular emerging trend in design such as the increasing size of a 'front grill'. Additionally, including information about consumers' demographic and behavioral profiles, such as age group, vehicle ownership history, purchase patterns, and design preferences, could provide opportunities for designers to develop robust hypotheses that help close the gap between their own views and those of consumers. Similar to the value that designers saw in using concrete design examples in interpolation images to aid communication, we believe that increasing data transparency through interactive visualizations can minimize the uncertainty that designers experience in design time.

8 CONCLUSION

We developed a novel framework and system that combines machine learning, human aesthetic assessments, and visualization. We hope our case study stimulates discussions in the HCI community around using this framework for professional product design practice.

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A TECHNICAL IMPLEMENTATION

A.1 Image Pre-processing

We applied various computational image editing techniques such as image masking and other pre-processing steps to prepare our image dataset, beginning with cropping wheel covers from a set of 820 images⁶ of driver-side views of vehicle models produced in

⁶The images used in this study are the property of EVOX Productions, LLC and are subject to copyright law. The appropriate licenses and permissions have been obtained to ensure the rightful use of these images in our study. For more information, visit <https://www.evostock.com/>.

2022. To remove any bias that can come from brand awareness, we removed logos displayed on the center hub cap. We used an open source tool <http://legacy.imagemagick.org/> to execute cropping and logo removal in batch. To maintain visual consistency, we manually removed wheel images that contained overly vibrant colors such as brake caliper covers from particular sports car models. Lastly, after removing any duplicate images, we ended with a final dataset of 686 wheel images.

A.2 Computing Interpolations

We generated all interpolations across 19,900 combinations of unique image pairs with compute jobs distributed across a total of 10 GPUs on AWS EC2 g5.xlarge g5 and g5.2xlarge instances. Each interpolation, comprising 15 images, were generated at a rate of 42 per hour or approximately 1000 per day. We used the implementation of ControlNet[21] and the following input parameters to compute the wheel interpolations:

```
prompt = 'wheel, high resolution'
object = "wheel"
qc_prompt = 'wheel, high resolution'
negative_prompt = 'cropped, worst quality, low quality, unfocused, printed words, pixelated, overexposed, oversaturated, cartoon, low saturation, jpeg artifacts, disproportioned, low res, watermark, text, ugly, morbid, mutilated, mutation, deformed, blurry, bad proportions, disfigured, deformed parts, error'
qc_neg_prompt = 'low quality'
ddim_steps = 200
ddim_eta = 0.1
num_frames = 17
guide_scale = 6
min_steps = 0.35
max_steps = 0.65
optimize_cond = 1
n_choices = 1
pose = 'canny'
```

B DATA COLLECTION FOR HUMAN AESTHETIC ASSESSMENTS

B.1 Computing CLIP Scores of Style Keywords

Because we did not know how the general public will interpret an image in association with a particular style keyword, we used CLIP [15] to measure an image's proximity to text. We encoded each pair of text and image to embeddings using CLIP and computed the cosine similarity as the CLIP scores for each of the 25 style keywords ([w]) identified in the formative study (Section 3) with respect to the 200 wheel images we sampled in the dataset. To be inclusive of the possible contexts in which LLM's are trained, we computed CLIP scores separately for the following inputs in the object parameter: [w], [w]+wheels, [w]+car+wheels. We then averaged the CLIP scores across the 3 different conditions to arrive at a final list of keywords and matching CLIP scores.

[w]	CLIP score				
Polished	.276	Rugged	.267	Classic	.258
Futuristic	.274	Sculpted	.266	Elegant	.252
Modern	.272	Efficient	.266	Innovative	.252
Dynamic	.271	Aggressive	.262	Tough	.251
Aerodynamic	.270	Luxury	.262	Chic	.251
Complex	.270	Emotional	.262	Sporty	.248
Refined	.270	Bold	.260	Strong	.247
Sleek	.269	Streamlined	.260		
Contemporary	.269	Stylish	.260		

Table 2: Average CLIP scores (in descending order) computed for 25 style keywords identified in the formative study. Final selection of nine style keywords are bolded.



Please rate the following questions
[PRACTICE]

Dynamic

Not at all 0 1 2 3 4 5 Extremely 6

Luxury

Not at all 0 1 2 3 4 5 Extremely 6

Polished

Not at all 0 1 2 3 4 5 Extremely 6

Modern

Not at all 0 1 2 3 4 5 Extremely 6

Rugged

Not at all 0 1 2 3 4 5 Extremely 6

Futuristic

Not at all 0 1 2 3 4 5 Extremely 6

Sculpted

Not at all 0 1 2 3 4 5 Extremely 6

Complex

Not at all 0 1 2 3 4 5 Extremely 6

Bold

Not at all 0 1 2 3 4 5 Extremely 6

Figure 4: Example question from the human aesthetic assessments survey.

From this list, we filtered out keywords with the lowest scores as well as ones that are semantically similar (e.g. modern vs contemporary) to each other. ‘Emotional’ was removed due to its strong association with the ‘emotional wheel chart’ by Psychologist Robert Plutchik [14]. Finally, a senior-level exterior designer helped identify the final list of nine style keywords, which we used in the survey to collect human aesthetic assessments. Table 2 shows average CLIP scores computed for each keyword.

B.2 Data Collection

Using the nine style keywords identified from above, we deployed a Qualtrics survey on the Prolific platform. 160 people were recruited to provide Likert ratings for 200 wheel images on a 7-point scale, for all nine style keywords per image (Fig. 4). Each respondent rated 30 different wheel images in the allotted time of 30 minutes.

ID	Role	YoE
p_f^1	Senior Creative Designer	8
p_f^2	Senior Lead Designer	22
p_f^3	Senior Lead Designer	13
p_f^4	Senior Lead Designer	26
p_f^5	Senior Creative Designer	18

Table 3: Formative study participants

ID	Role	YoE
p_u^1	Automotive Designer	20
p_u^2	MFA student	0.5
p_u^3	MFA student	0.5
p_u^4	MFA student	3
p_u^5	Automotive Designer	2
p_u^6	Automotive Designer	6
p_u^7	MFA student	1
p_u^8	MFA student	1
p_u^9	MFA student	2.5
p_u^{10}	MFA student	2

Table 4: User study participants. For students, years of experience includes internships and any professional experience outside of training.

C PARTICIPANT DEMOGRAPHICS

Table 4 shows the role and years of experience of 10 participants.